SECURITY ENGINEERING RESEARCH POSTER

Evaluating an Offline Detector Against Adversarial Prompt Attacks

A Reproducible Framework for Prompt-Attack Generation and Defensive Evaluation

github.com/poojakira/llm-redteam-framework - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

182	1	94.22%	1.00
Tests passed	Tests skipped	Statement coverage	Random-split F1

SYSTEM VIEW

Architecture and Threat-Control Flow

Trust Boundary

Input surface
Data, models, requests, policies, or artifacts entering the tested path.

Control surface
Repository-specific validation, analysis, and fail-closed checks.

Evidence surface
Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01 **Generate adversarial and benign prompt corpora across defined attack categories.** →

02 **Train/evaluate the offline TF-IDF + Logistic Regression detector.** ↓

03 **Compare random, grouped-template, structural-fixture, and novel-phrasing OOD evaluations.** →

04 **Emit JSON/SARIF evidence for CI.** ↓

05 **Keep live-model behavior claims out of scope unless separately measured.**

Research Contribution

A reproducible offline prompt-attack evaluation framework that reports both the stronger in-distribution result and the weaker novel-phrasing result, making the detector's generalization gap explicit instead of hiding it behind a random split.

Verified Evidence

EV-01	Grouped-template F1: 0.9714	EV-02	Novel-phrasing OOD F1: 0.7188
EV-03	OOD precision: 0.5897	EV-04	OOD recall: 0.92
EV-05	OOD false-positive rate: 64% (16/25)	EV-06	182 passed, 1 skipped.
EV-07	94.22% statement coverage; CI gate is 90%.	EV-08	Current benchmark values remain pinned:
EV-09	random split F1 1.00	EV-10	grouped-template F1 0.9714

Method and Control Path

01 **Generate adversarial and benign prompt corpora across defined attack categories.**

02 **Train/evaluate the offline TF-IDF + Logistic Regression detector.**

03 **Compare random, grouped-template, structural-fixture, and novel-phrasing OOD evaluations.**

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Limitations and Residual Risk

- Offline detector evaluation does not prove whether a live LLM would follow an injected instruction.
- English/template coverage is finite.
- Structural fixtures are not equivalent to true OOD natural-language paraphrases.
- No real-world jailbreak-defense rate is claimed.

Reproducibility

```
Expected CI evidence: 182 passed, 1 skipped, 94.22% coverage.
```

Evidence Provenance

Claim discipline
Metrics come from the repository's verified metrics and research brief.

Scope discipline
Benchmarks are reported only at their documented fixture, dataset, or test scope.

Reproduction
PDF and PNG are generated locally from committed evidence without paid services.

Evidence-backed security engineering

README, verified metrics, tests, CI, and committed artifacts remain authoritative. Poster layout emphasizes architecture, controls, evidence, limitations, and reproduction rather than unsupported efficacy claims.

36 x 48 in
PDF + PNG parity